340061 – Introduction to Graphics and Engineering Design

Winter Semester, 2026

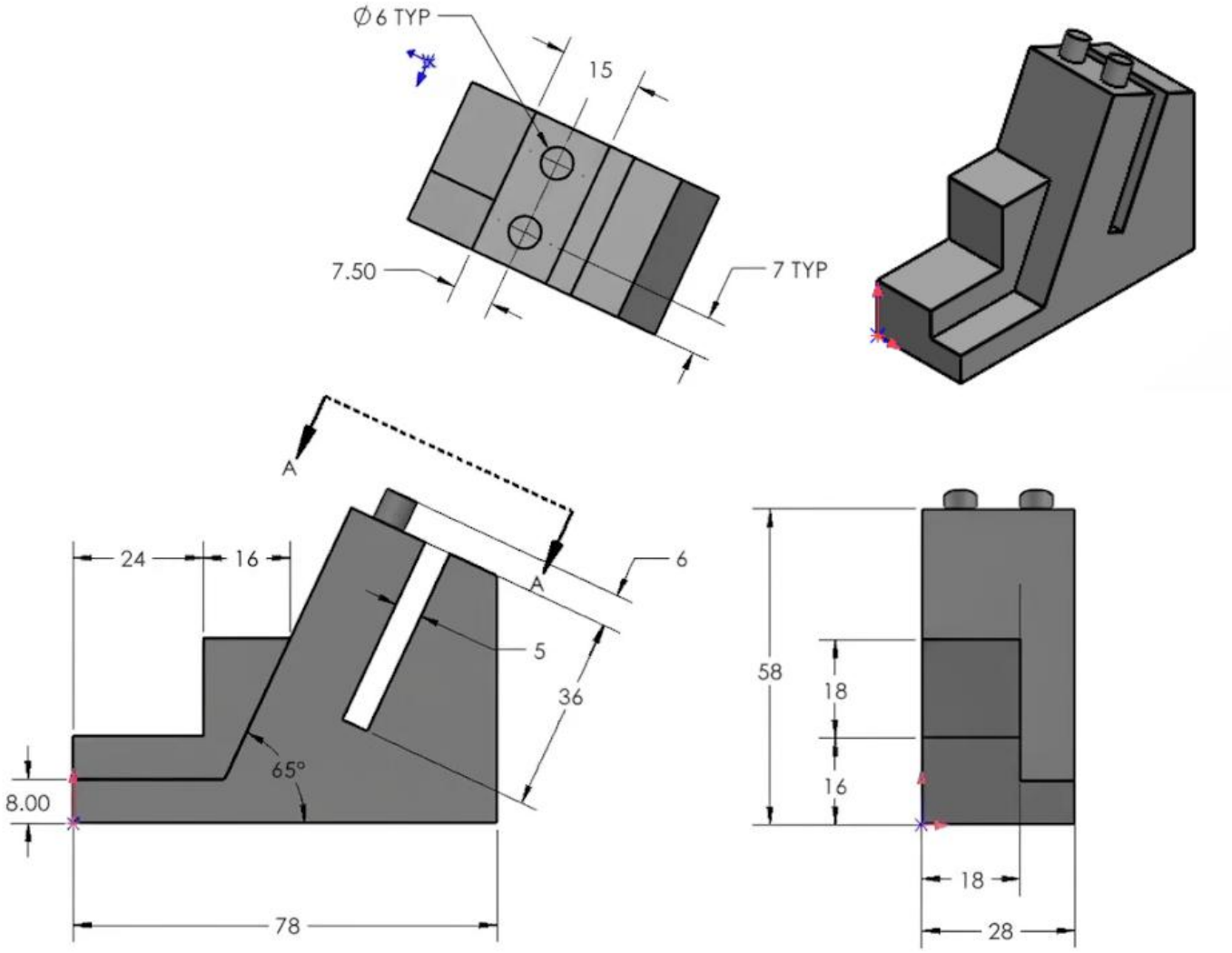
Lab 5 Assignment
Basic Part Modeling

General Instructions

1. This assignment is done in **pairs** and submitted through Moodle. Only students who belong to a pair group can submit.
2. Late submissions will **not** be accepted.
3. Pay attention to **file names** as instructed in the questions, E.G.: "**Q1.SLDPRT**"
   Incorrect names will result in **point penalties**.
4. Submit all files as a single compressed **.ZIP** file, other formats will not be accepted. Verify your files open properly in SolidWorks 2026.
5. Your solutions must follow the rules and principles learned in class.
   E.G.: Fully Defined sketches, Design Intent, correct dimensions, relations and features.
6. **AI and Plagiarism** - Do not use AI to automate, generate or complete any part of this assignment. Do not share or copy files from other students. All students involved in a plagiarism case (the students who copied and the students whose work was copied) will be given zero on the assignment.

Q1.sldprt

Model the following part:



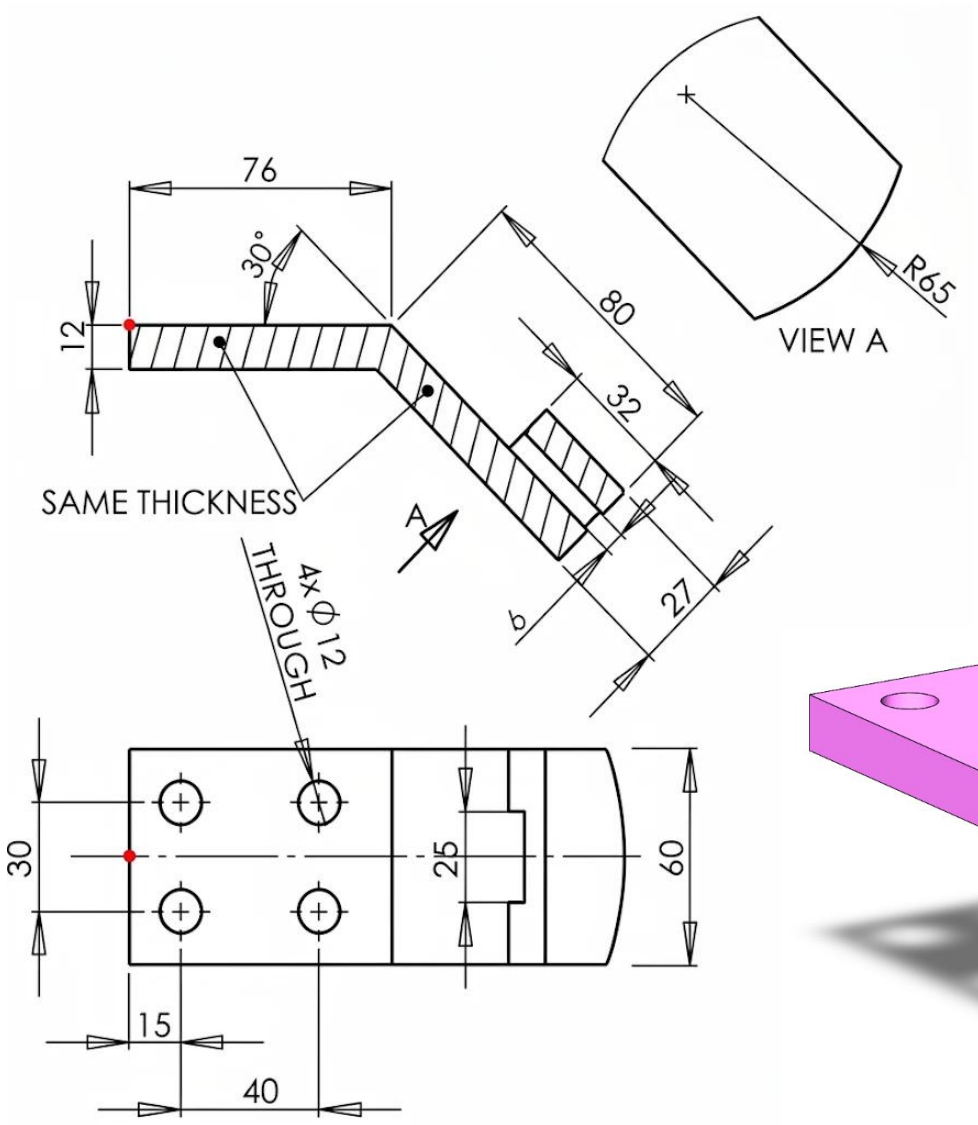
- Dimensions are in millimeters.
- Design the part so that the main view is the **front plane.**
- Reference model volume is $70039 \pm 1\%\ [mm^3]$

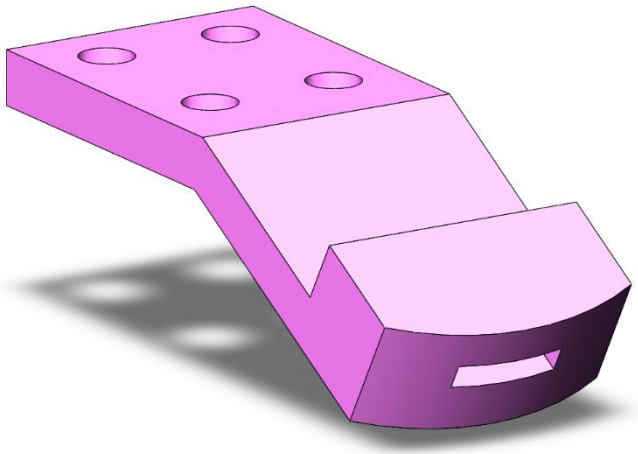
Q2.sldprt

Submit the part **Q2.sldprt** that you designed in Lab 4.

Q3.sldprt

Model the following part:





- Design the part so that the symmetry plane is the **front plane**. Place the origin at the location of the red dot.
- Make sure all features and sketches are **fully defined.**
- Verify design intent.
- For reference, the resulting model volume is $124768 \pm 1\%\ [mm^3]$

Q4.sldprt

Create a new part and model the following geometry according to the given views.

- Design the part so that the symmetry plane is the **front plane**. Place the origin in the center of the R21 hole.
- Dimensions are in millimeters. Use all the shown dimensions, do not add more dimensions.
- Use correct sketch relations.
- Verify design intent. For example, upon adjusting the 48[mm] width the part must remain symmetric around the front plane.
- Make sure all features and sketches are **fully defined**!
- For reference, the resulting model volume is $201961 \pm 1\%\ [mm^3]$

